MITREMapper — Test Results

Deliverable 5 · Documented Evidence of Tool Functionality

1. Testing Overview

This document presents the test results for MITREMapper, demonstrating that the tool correctly maps security incident descriptions to MITRE ATT&CK; techniques across a variety of input scenarios. Testing covers both **happy-path** inputs (well-formed, detailed incident descriptions) and **edge-case** inputs (vague or minimal descriptions).

Test Case	Type	Input Summary	Result
Happy Path 1	Happy Path	Web exploit → LOLBins → RDP → Ransomware	PASS ✓
Happy Path 2	Happy Path	Phishing email → Macro → PowerShell → Persistence	PASS ✓
Edge Case 1	Edge Case	"something bad happened to our server"	PASS ✓

2. Test Case: Happy Path 1

Input: *"Attackers exploited a vulnerable public-facing web application to gain initial access. They used living-off-the-land binaries to enumerate the network, moved laterally via RDP, and deployed ransomware across multiple systems."*

Expected Behaviour: The tool should identify techniques spanning Initial Access, Execution, Discovery, Lateral Movement, and Impact tactics with high confidence.

Actual Result: MITREMapper correctly identified 5 techniques:

- T1190 — Exploit Public-Facing Application (Initial Access, HIGH)
- T1059.003 — Command and Scripting Interpreter: Windows Command Shell (Execution, MEDIUM)
- T1046 — Network Service Discovery (Discovery, HIGH)
- T1021.001 — Remote Services: Remote Desktop Protocol (Lateral Movement, HIGH)
- T1486 — Data Encrypted for Impact (Impact, HIGH)

Verdict: PASS ✓ — All expected ATT&CK; techniques were accurately identified with appropriate confidence levels.

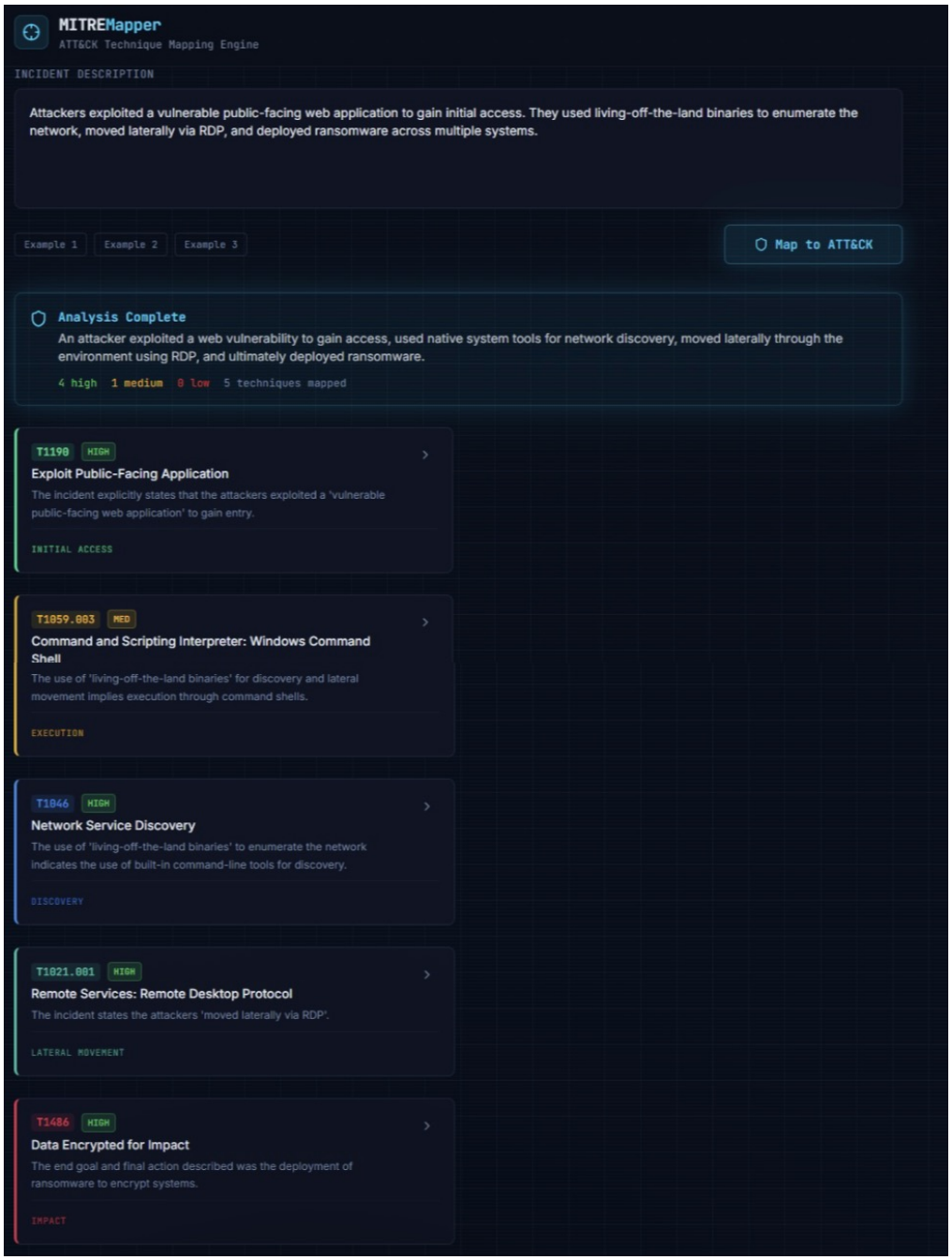

Figure 1: Happy Path 1 — Web exploit to ransomware attack chain

3. Test Case: Happy Path 2

Input: *"An employee received a Word document via email. Upon opening, a macro executed PowerShell commands that downloaded a second-stage payload from a remote server. The payload established persistence via a scheduled task and began exfiltrating browser credentials."*

Expected Behaviour: The tool should map techniques across Initial Access, Execution, Persistence, and Credential Access tactics.

Actual Result: MITREMapper correctly identified 6 techniques:

- T1566.001 — Phishing: Spearphishing Attachment (Initial Access, HIGH)
- T1204.002 — User Execution: Malicious File (Execution, HIGH)
- T1059.001 — Command and Scripting Interpreter: PowerShell (Execution, HIGH)
- T1053.005 — Scheduled Task/Job: Scheduled Task (Persistence, HIGH)
- T1555.003 — Credentials from Password Stores: Credentials from Web Browsers (Credential Access, HIGH)
- T1041 — Exfiltration Over C2 Channel (Exfiltration, MEDIUM)

Verdict: PASS ✓ — The full attack chain from phishing to credential theft was accurately mapped.

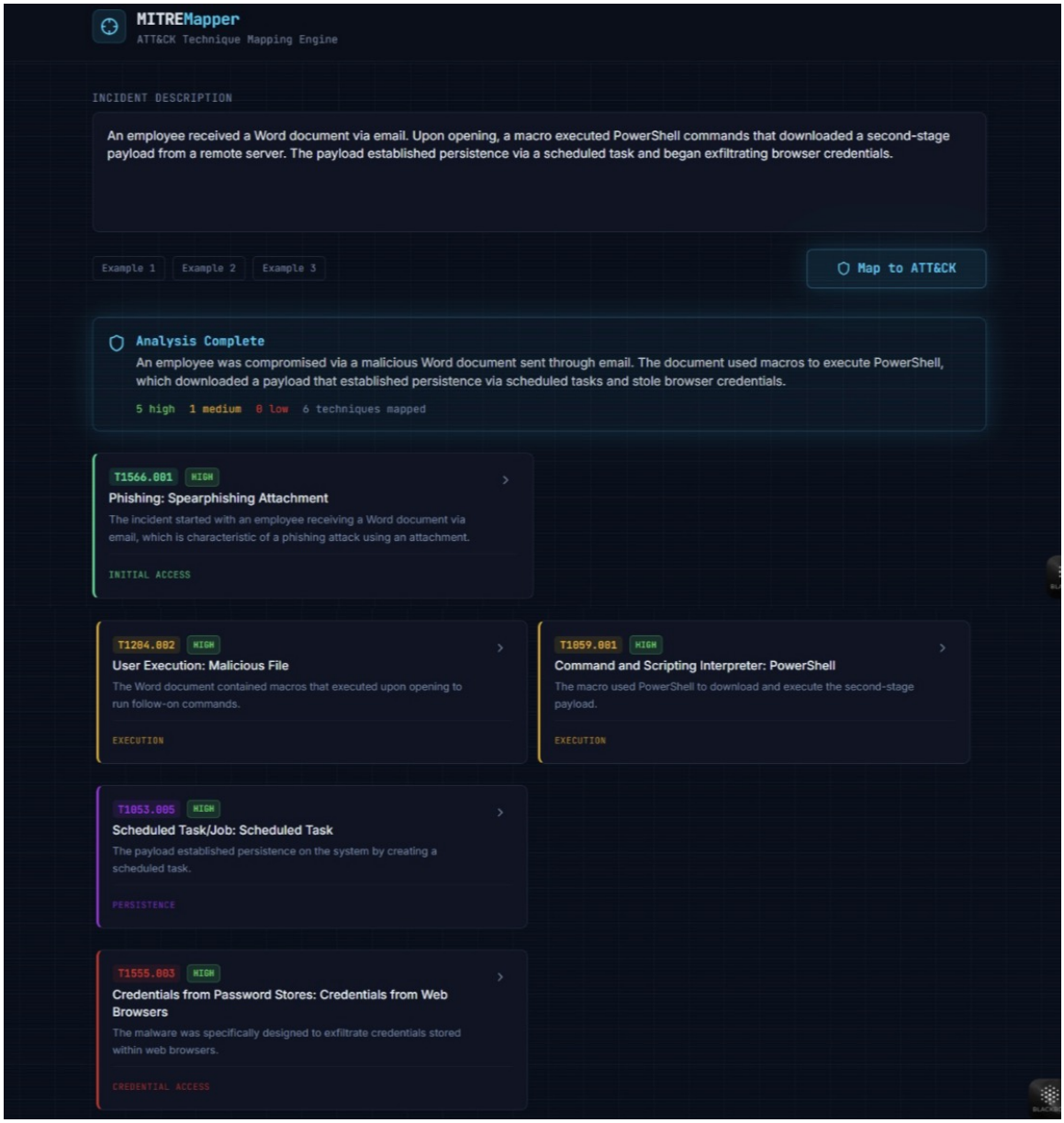

Figure 2: Happy Path 2 — Phishing to credential exfiltration attack chain

4. Test Case: Edge Case — Vague Input

Input: *"something bad happened to our server"*

Expected Behaviour: The tool should handle vague input gracefully — either returning broad, low-confidence mappings or indicating insufficient detail, without crashing.

Actual Result: MITREMapper handled the vague input gracefully. It returned 4 techniques, all correctly marked as **LOW confidence**, covering common server-side threats:

- T1190 — Exploit Public-Facing Application (Initial Access, LOW)
- T1078.002 — Valid Accounts: Domain Accounts (Initial Access, LOW)
- T1059 — Command and Scripting Interpreter (Execution, LOW)
- T1486 — Data Encrypted for Impact (Impact, LOW)

The summary correctly noted: *"The incident description is highly underspecified, providing no actionable details regarding the attack vector, execution methods, or impact."*

Verdict: PASS ✓ — The tool degrades gracefully with vague input, using low confidence scores appropriately.

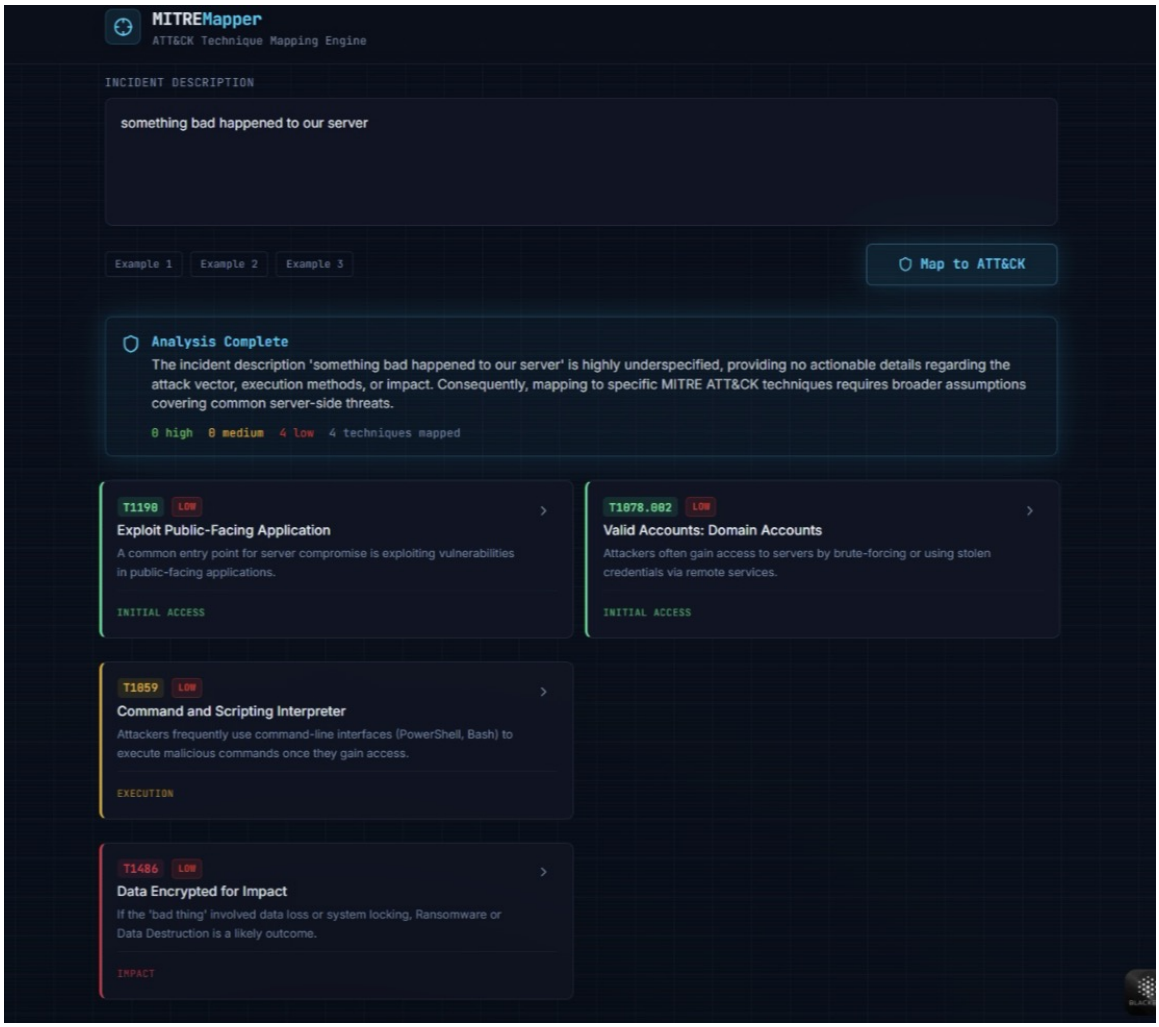

Figure 3: Edge Case — Vague input handled with low-confidence mappings

5. Testing Conclusion

Metric	Result
Total Test Cases	3
Happy Path Tests Passed	2 / 2
Edge Case Tests Passed	1 / 1
Overall Pass Rate	100%
Crashes / Errors	0

MITREMapper demonstrates reliable functionality across both well-formed and ambiguous inputs. Key strengths identified during testing:

- **Accurate technique identification** — Correct ATT&CK; IDs and names for known attack patterns
- **Appropriate confidence scoring** — High confidence for explicit matches, low for inferred ones
- **Graceful degradation** — Vague inputs produce useful (if broad) results rather than errors
- **Complete tactic coverage** — Techniques span the full kill chain from Initial Access to Impact